



Git Command Roadmap

Complete guide to Git commands with simple explanations

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Core

`git init`

Initialize a new Git repository

`git clone`

Clone a repository from a remote source

`git add`

Stage changes for the next commit

`git commit`

Save staged changes with a message

`git status`

Show the working tree status

`git diff`

Show changes between commits or working tree

`git checkout`

Switch branches or restore files

`git reset`

Undo changes or unstage files

`git log`

Show commit history

`git show`

Show details of a specific commit

`git tag`

Create, list, or delete tags

`git push`

Upload local commits to remote repository

`git pull`

Fetch and merge changes from remote



Branching

`git branch`

List, create, or delete branches

`git checkout -b`

Create and switch to a new branch

`git merge`

Combine branches together

`git rebase`

Reapply commits on top of another branch

`git branch --set-upstream-to`

Set upstream tracking for a branch

git branch --unset-upstream

Remove upstream tracking

git cherry-pick

Apply specific commits to current branch



Merging

git merge

Join two or more development histories

git rebase

Rewrite commit history by moving commits



Stashing

git stash

Temporarily save uncommitted changes

git stash pop

Apply and remove the latest stash

git stash list

Show all saved stashes

git stash apply

Apply a stash without removing it

git stash drop

Delete a specific stash



Remotes

`git remote`

Manage remote repository connections

`git remote add`

Add a new remote repository

`git remote remove`

Remove a remote connection

`git fetch`

Download objects from remote repository

`git pull`

Fetch and integrate remote changes

`git push`

Update remote refs with local commits

`git clone --mirror`

Create a bare mirror clone of repository



Configuration

`git config`

Set or get repository configuration

`git config --global`

Configure Git settings globally

git config --unset

Remove a configuration setting

<> Plumbing (Low-Level)

git cat-file

Display content of repository objects

git checkout-index

Copy files from index to working tree

git commit-tree

Create a new commit object

git diff-tree

Compare trees in repository

git for-each-ref

Iterate over all refs

git hash-object

Compute object ID and store object

git ls-files

Show information about files in index

git ls-remote

List references in remote repository

git merge-tree

Show three-way merge without touching index

git read-tree

Read tree information into index

git rev-parse

Parse revision parameters

git show-branch

Show branches and their commits

git show-ref

List references in local repository

git symbolic-ref

Read or modify symbolic references

git update-ref

Update ref safely with old value check



Porcelain (High-Level)

git blame

Show who last modified each line

git bisect

Binary search to find bug-introducing commit

git checkout

Switch branches or restore files

git commit

Record changes to repository

git diff

Show changes between commits

git fetch

Download objects from another repository

git grep

Search for patterns in tracked files

git log

Show commit logs

git merge

Join development histories

git push

Update remote repository

git rebase

Reapply commits on different base

git reset

Reset current HEAD to specified state

git show

Show various types of objects

git tag

Create or manage tags



Alias

```
git config --global alias.<name> <command>
```

Create custom command shortcuts



Hook

```
git config --local core.hooksPath <path>
```

Set custom Git hooks directory path



Experimental / Advanced

```
git annex
```

Manage large files with Git

```
git am
```

Apply patches from mailbox

```
git cherry-pick --upstream
```

Cherry-pick with upstream option

```
git describe
```

Give object a human-readable name

```
git format-patch
```

Prepare patches for email submission

```
git fsck
```

Verify connectivity and validity of objects

```
git gc
```

Cleanup unnecessary files and optimize

git help

Display help information

git log --merges

Show only merge commits

git log --oneline

Show condensed commit history

git log --stat

Show commit stats with file changes

git mergetool

Run merge conflict resolution tool

git mv

Move or rename files

git prune

Remove unreachable objects

git pull --rebase

Pull and rebase instead of merge

git push --mirror

Push all refs to remote

git push --tags

Push all tags to remote

git reflog

Show reference logs

git replace

Create replacement objects

git reset --hard

Reset and discard all changes

git reset --mixed

Reset keeping changes in working directory

git revert

Create new commit that undoes changes

git rm

Remove files from working tree and index

git stash save

Save work in progress with message

git subtree

Manage project dependencies as subtrees

git tag --delete

Delete a tag

git tag --sign

Create signed tag

git worktree

Manage multiple working trees

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